

Introductory Microbiology

Kingdom Monera

Characteristics

- simplest, microscopic, prokaryotic unicellular organisms
- Devoid of true nucleus and membrane bound cell organelles like plastids, golgibodies, mitochondria, etc.
- Most of them are saprophytes and parasites but, few are photoautotrophs.
- Circular DNA without histone protein
- Ribosome is 70s type
- Possess rigid cell wall made up of peptidoglycan or murein [absent in mycoplasma]
- Autotrophic or heterotrophic mode of nutrition

Bacteria

- the simplest and the smallest microscopic organisms with cellular structure.
- Ubiquitous (found everywhere)
- prokaryotic type of cell structure.
- lack true or advanced nucleus. Nucleoid (incipient nucleus or genophore) is present without nuclear membrane, nucleoplasm, nucleolus and chromatin. Nucleoid consists of DNA, RNA and protein. DNA is circular without histone protein.
- Membrane bound cell organelles like Golgi body, mitochondria, plastid, endoplasmic reticulum, etc. are absent.
- 70s types of ribosomes are found in each cell
- cell wall is made up murein (mucopeptide or peptidoglycan) that consists of N-acetyl glucosamine and N-acetyl muramic acid instead of cellulose.
- autotrophic or heterotrophic mode of nutrition.
- known as decomposers and mineralisers in the biosphere.
- True vacuole is absent but gas vacuoles may be present in some aquatic bacteria.
- reserved food material is glycogen or fats.
- Flagella, if present, are single stranded and consist of protein called flagellin.
- Cell division takes place by amitosis.
- Reproduction commonly takes place by the binary fission.
- sexual reproduction is absent but genetic recombination takes place by conjugation, transformation and transduction in some bacteria.
- *Vibrio*, *Diplococcus*, *Bacillus*, *Streptococcus*, etc. are examples of bacteria.

- **Classification:**

Cohn (1872) classified bacteria into following types on the basis of shape of the cell.

1. Coccus (plural-cocci) bacteria:

- spherical, oval or rounded in shape and present in following different forms:

- a. Monococcus: It is oval or rounded bacterium present in single.e.g. *Micrococcus luteus*
- b. Diplococci: They are oval or rounded bacteria present in pairs.e.g. *Neisseria spp.*
- c. Tetrads: They are oval or rounded bacteria present in tetrad.e.g. *Gaffkya*
- d. Streptococci: They are oval or rounded bacteria present in chain like form.e.g. *Streptococcus pyogenes*
- e. Staphylococci: They are oval or rounded bacteria present in grapes like irregular cluster.e.g. *Staphylococcus aureus*
- f. Sarcina: They are oval or rounded bacteria present in box like cubical pockets.e.g. *Sarcina*

2. Bacillus (Plural-Bacilli) bacteria:

- most common type of bacteria.
- cylindrical or rod shaped structure.
- They are present in following forms:

- a. Monobacillus: bacterium present in single. *Salmonella enterica*
- b. Diplobacilli: bacteria present in pairs.e.g. *Moraxella bovis*
- c. Streptobacilli: bacteria present in chains.e.g. *Streptobacillus*
- d. Palisade: bacteria present in stacks like form.(Chinese letter like). Eg. *Corynebacterium diphtheriae*

3. Helical bacteria:

- coiled bacteria and are of two types:

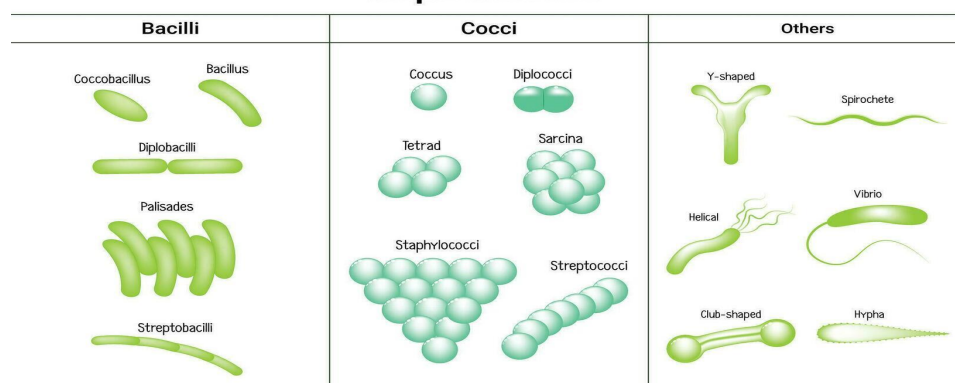
- a. Spirillum (plural-spirilla) type: spirally coiled or twisted bacteria e.g. *Spirillum*.

- b. Vibrio (plural-vibrios) type:

- comma or C-shaped bacteria. e.g. *Vibrio spp*

4. pleomorphic: irregular in shape E.g. *Azotobacter spp*

Shapes of Bacteria



Cell structure:

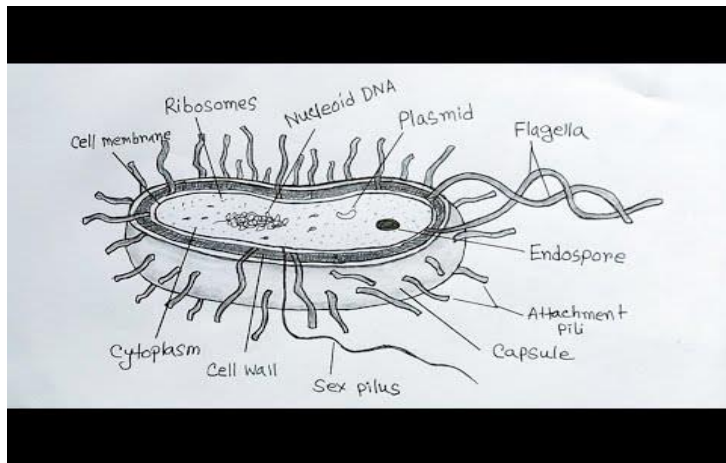
The bacterial cell consists following components:

1. Cell wall:

- Outermost rigid, protective covering of the cell which is made up of peptidoglycon or murein which consists of N-acetyl glucosamine(NAG), N-acetyl muramic acid (NAM) and Diaminopimelic acid.
- It gives the cell its shape and surrounds the cytoplasmic membrane, protecting it from the environment.
- Based on their wall structure and staining ability with the Gram's stain, bacteria are classified into following two types: Gram positive and Gram negative bacteria.

Difference between Gram positive and Gram negative bacteria

SN	Gram positive bacteria	Gram negative bacteria
1.	Cell wall is thick (15-20 nm) and homogenous.	Cell wall is thin (7-12 nm) and heterogenous
2.	They retain purple colour after Gram staining	They retain red/pink colour after Gram staining
3.	large amount of peptidoglycan (70-80%) in cell wall.	less amount of peptidoglycan (3-12%) in cell wall.
4.	Techoic acids are present.	Techoic acids are absent.
5.	Diaminopimelic acid is absent in cell wall.	Diaminopimelic acid is present in cell wall.
6.	Fimbriae/ sex pilli are absent.	Fimbriae/ sex pilli are present.
7.	Mesosomes are present.	Mesosomes are absent.
8.	Generally non-virulent and non-pathogenic	Generally virulent and pathogenic.
9.	They produce exotoxin.	They produce endotoxin.
10.	Mostly coccus and non-capsulated.	Mostly bacillus and capsulated
11.	E.g. <i>Staphylococcus</i> , <i>Streptococcus</i> , <i>Clostridium</i> , etc.	E.g. <i>E.coli</i> , <i>Vibrio</i> , <i>Salmonella</i> , <i>Pseudomonas</i> , etc.



2. Slime layer or capsule

- covered externally either slime layer or capsule.

a. Slime layer: In some bacteria, the cell is externally covered by a thin extra gelatinous viscous layer which is made up of polysaccharides called slime layer.

b. Capsule: In some virulent bacteria, cell wall is covered by thick extra-gelatinous mass which is made up of polysaccharides and amino acids called capsule.

- The slime layer or capsule protects the bacterial cell from desiccation, antibiotics, antibodies, etc.

3. Cell membrane:

- thin and elastic membrane present inner to the cell wall which is made up of phospholipids, proteins and polysaccharides.

- In Gram +ve bacteria, mesosomes or chondroids are present. Mesosomes are small circular or semicircular bodies formed by the infolding of cell membrane.

- It consists of several respiratory enzymes which help in respiration.

4. Cytoplasm:

- a granular, colourless fluid mass present between the cell membrane and nuclear body.

- consists of ribosomes, gas vacuoles, soluble proteins, granules of fat, lipids, etc. Membrane bound cell organelles like mitochondria, plastids, endoplasmic reticulum, Golgi body, etc. are absent.

- Important cytoplasmic bodies present in cytoplasm are as follows:

a. Gas vacuoles: Gas vacuoles are present in the aquatic bacteria that help in flotation.

b. Reserve food material: The reserve food materials are glycogen, volutin granules, polyhydroxybutyric acid, lipids or proteins.

c. Thylakoids or lamellae: Some photosynthetic bacteria consist of number of thylakoids which are embedded in the cytoplasm called chromatophores. The photosynthetic pigments (bacteriochlorophyll and chlorobium chlorophyll) are found in the membrane of thylakoids.

d. Ribosomes: Ribosomes are 70s (50s and 30s) type which are scattered in the cytoplasm. Sometimes smaller and larger subunits of ribosomes join with mRNA and form chain like polyribosomes. They help in protein synthesis.

5. Incipient nucleus or nuclear body:

- The bacterial cell lack true organised nucleus.
- Nucleus is present without nuclear membrane, nucleoplasm and nucleolus.
- It consists a double stranded circular DNA molecule without histone protein. Such type of nucleus is called incipient nucleus, genophore or nucleoid.
- Plasmid: Many bacterial cells consist of extra-chromosomal self replicating circular DNA in addition to nuclear DNA called plasmid. Sometimes, plasmid may be integrated to incipient nucleus and forms episome. Episome helps in transfer of genetic material between the different bacteria at the time of conjugation.

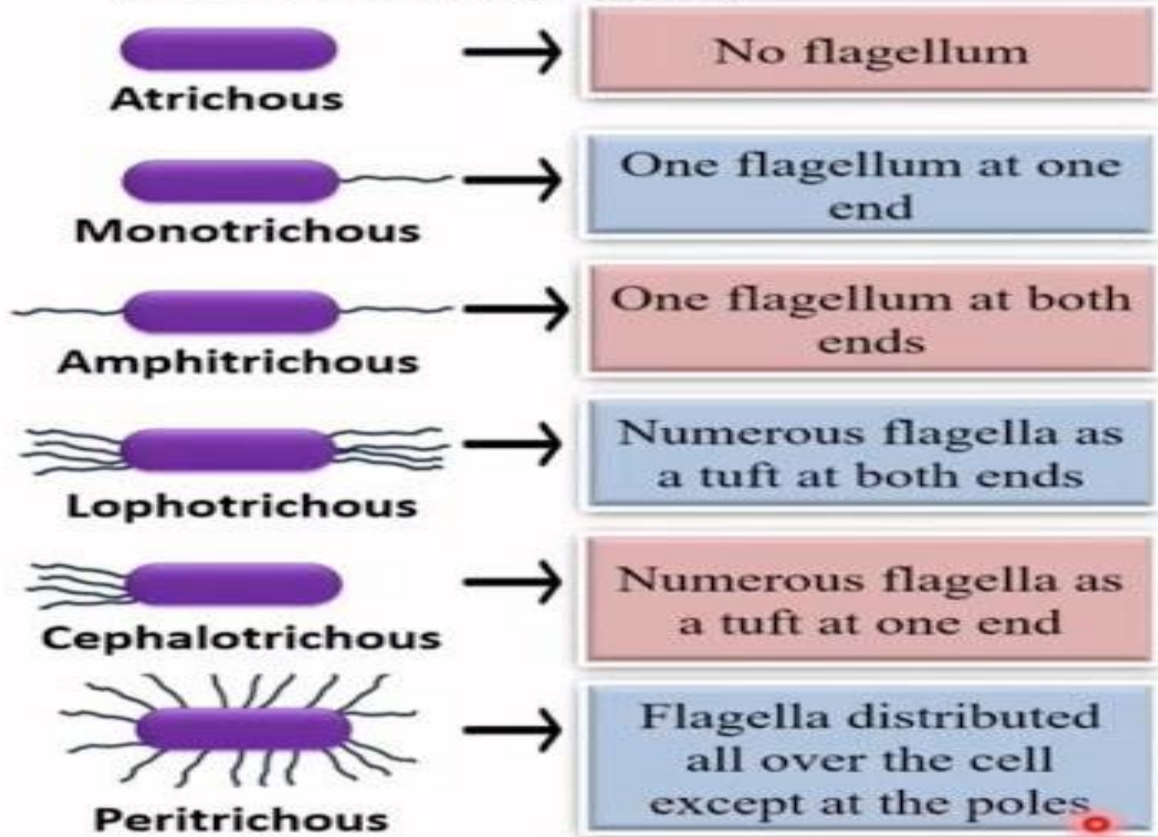
6. Sex pilli or fimbriae:

- fine, short straight rigid hair like protoplasmic outgrowths developed from the cell wall of gram negative bacteria.
- made up of protein called pilin.
- help in attachment and formation of conjugation tube during conjugation.

7. Flagella:

- long whips or hair like protoplasmic thread found in bacillus and helical type of bacteria and completely absent in coccus type of bacteria.
- made up of protein called flagellin.
- Each flagellum is made up of 3 parts: filament, hook and basal body.
- The basal body of flagella are attached to the cell membrane by a cytoplasmic granule called blepharoplast.
- Flagella help in locomotion or movement.

Types of Bacteria (Based on Flagellum)



• Depending upon the flagellation, bacteria are following types atrichous and trichous bacteria:

a. Atrichous bacteria: Flagella are absent e.g. *Lactobacillus*

b. Trichous bacteria: Flagella are present in bacterial cell.

• Monotrichous: Single flagellum at one end of the cell e.g. *Vibrio*

• Amphitrichous: One flagellum at each end of the cell e.g. *Nitrosomonas*

- Cephalotrichous: A group of two or more flagella present at only one end of the cell e.g. *Pseudomonas*
- Lophotrichous: A cluster of flagella at both ends of the cell e.g. *Spirillum*
 - peritrichous: tuft of flagella projecting in all directions. e.g. *Salmonella Typhi*

Mode of Nutrition in Bacteria

- Bacteria mainly show heterotrophic mode of nutrition. But some bacteria show autotrophic mode of nutrition.
 1. Autotrophic bacteria: Autotrophic bacteria can synthesize their own food either by the process of bacterial photosynthesis or by the process of chemosynthesis. They are of two types:
 - A. Photoautotrophic Bacteria: Photoautotrophic bacteria synthesize their own food by utilisation of CO₂ as carbon source, H₂S thiosulphate, hydrogen or even from some other organic compounds (hydrogen/electron donors) in the presence of sunlight and bacteriochlorophyll or chlorobium chlorophyll. Photoautotrophic bacteria can be divided into following three types:
 - i. Green Sulphur bacteria: They synthesize their own food by utilization of CO₂ and H₂S in the presence of sunlight and chlorobium chlorophyll. Sulphur is produced. e.g. *Chlorobium*.

$$\text{CO}_2 + \text{H}_2\text{S} \rightarrow [\text{CH}_2\text{O}] + \text{S}$$
 - ii. Purple Sulphur bacteria: They synthesize their own food by utilization of carbon dioxide and thiosulphate compound in the presence of sunlight and bacteriochlorophyll e.g. *Chromatium*.

$$6\text{CO}_2 + 3\text{Na}_2\text{S}_2\text{O}_3 \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{H}_2\text{O} + 6\text{NaHSO}_4$$
 - iii. Purple non-Sulphur bacteria: They synthesize their own food by utilization of carbon dioxide and organic compounds in the presence of sunlight and bacteriochlorophyll. Sulphur is not formed as a byproduct e.g. *Rhodospirillum*.

$$6\text{CO}_2 + 12\text{CH}_3\text{CHOHCH}_3 \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{H}_2\text{O} + 12\text{CH}_3\text{COCH}_3$$
 - B. Chemosynthetic Bacteria:
 - Non-photosynthetic autotrophic bacteria.
 - Light is not involved in this process.
 - Utilize the chemical energy i.e. Oxidation of H₂S, NH₃, Fe²⁺, NO₂⁻ etc. releases energy to reduce CO₂ to organic food

Chemosynthetic Bacteria:

Bacteria	Substrate Oxidized	Energy Reaction
Nitrifying bacteria: <i>Nitrosomonas</i> <i>Nitrobacter</i>	Ammonia → Nitrite	$2\text{NH}_3 + 3\text{O}_2 \rightarrow 2\text{HNO}_2 + 2\text{H}_2\text{O} + \text{Energy}$
	Nitrite → Nitrate	$2\text{HNO}_2 + \text{O}_2 \rightarrow 2\text{HNO}_3 + \text{Energy}$
Sulphur bacteria: <i>Thiobacillus</i>	H ₂ S → Sulphate	$\text{H}_2\text{S} + 2\text{O}_2 \rightarrow \text{H}_2\text{SO}_4 + \text{Energy}$
Iron bacteria: <i>Ferrobacillus</i>	Fe ²⁺ → Fe ³⁺	$4\text{FeCO}_3 + \text{O}_2 + 6\text{H}_2\text{O} \rightarrow 4\text{Fe(OH)}_3 + 4\text{CO}_2 + \text{Energy}$

Bacteria	Substrate Oxidized	Energy Reaction
Hydrogen bacteria: <i>Hydrogenomonas</i>	$\text{H}_2 \rightarrow \text{H}_2\text{O}$	$2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O} + \text{Energy}$

3. Heterotrophic Bacteria:

i.Parasitic bacteria: obtains food from host body and may be pathogenic or non-pathogenic. *Mycobacterium tuberculosis* (TB), *Vibrio cholerae* (Cholera), etc.

ii.Saprophytic bacteria: depends on dead and decaying organic matters. *Bacillus subtilis*, *Pseudomonas*, etc.

iii. Symbiotic bacteria: associates with other living beings where both of them are useful to each other. Eg. Rhizobium are present in the nodules of leguminous plant root.

Reproduction in Bacteria:

Bacteria reproduce rapidly through both asexual (common) and sexual means:

Asexual Reproduction

- Binary Fission: The most common method of bacterial reproduction, where a single cell divides into two identical daughter cells.
- Spore Formation: Some bacteria form specialized spores as a survival strategy to endure unfavorable conditions. These spores can withstand harsh environments and later germinate into viable bacterial cells.

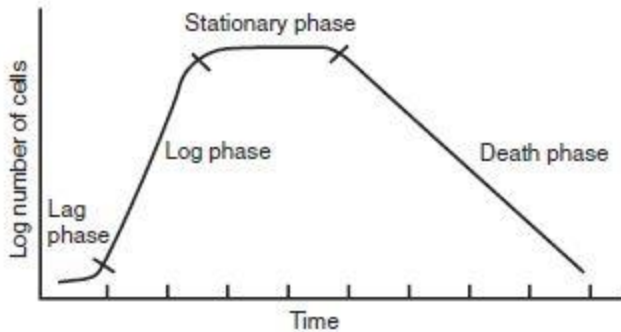
Sexual Reproduction

- Transformation: In transformation, bacteria can take up free DNA from their surroundings and incorporate it into their genome, leading to genetic diversity.
- Transduction: Transduction occurs when bacterial DNA is transferred from one cell to another by bacteriophages (viruses that infect bacteria).
- Conjugation: Conjugation involves the direct transfer of genetic material between two bacterial cells through a structure called the sex pilus.

Bacterial Growth

- Bacteria are unicellular organisms that tend to reproduce asexually by the means of binary fission. Bacterial growth is the increase in the number of bacterial cells rather than the increase in their cell size.
- The time taken for a bacterial cell to double is called generation time (doubling time). The generation time varies among different species of bacteria based on the environmental conditions they grow in. Generation time is typically about 20-30 minutes.

Phases of Bacterial Growth Curve



1. Lag Phase:

- The bacteria upon introduction into the nutrient medium take some time to adapt to the new environment.
- In this phase, the bacteria does not reproduce but prepares itself for reproduction. The cells are active metabolically and keep increasing in size. The cells synthesise RNA, growth factors and other molecules required for [cell division](#).

2. Log Phase (Exponential Phase):

- Soon after the lag phase, i.e., the preparation phase, the bacterial cells enter the log phase. The log phase is also known as the exponential phase.
- This phase is marked by the doubling of the bacterial cells. The log phase continues until there is depletion of nutrients. The stage also comes to a stop if toxic substances start to accumulate, resulting in a slower growth rate.

3. Stationary Phase:

- In the stationary phase, the rate of growth of the cells becomes equal to its rate of death. The rate of growth of the bacterial cells is limited by the accumulation of toxic compounds and also depletion of nutrients in the media. The cell population remains constant at this stage.

4. Death Phase (Decline Phase):

- This is the last phase of the bacterial growth. At this stage, the rate of death is greater than the rate of formation of new cells. Lack of nutrients, physical conditions or other injuries to the cell leads to death of the cells.

Economic importance

Beneficial activities :-

- Some species of bacteria like *Rhizobium*, *Chlostridium*, *Azotobacter*, fix atmospheric nitrogen in usable forms for plants.
- Some species of *Clostridium* are used in manufacturing of Vitamin B(riboflavin).
- Lactic acid bacteria(*Streptococcus lactis*) help in coagulation of milk.
- Three types of medicine – antibiotics, antisera and vaccines are formed by bacteria.
- Bacteria(*E. coli*) present in human colon synthesize vitamin B and releases it form human use.
- Bacteria help in digestion in herbivores animals in the digestion of cellulose. *Bacillus subtilis* produces the enzymes amylase and protease.

- g. By the activity of *Clostridium* the butyl alcohol and acetone are obtained from sugar molasses.
- h. It helps in fibers retting of jute, coconut and other fibrous plants.
- i. Bacteria like *Bacillus subtilis*, *Bacillus ramosus* etc. decompose complex organic compounds like proteins into ammonium compound. These are also known as putrefying bacteria.

Harmful activities :-

- a. Many human diseases such as diarrhoea, meningitis, pneumonia etc. are caused by pathogenic bacteria.
- b. Some denitrifying bacteria reduce the soil fertility by depleting the nitrogen contents of soil.
- c. Some bacteria spoil food stuffs, leather and wooden articles.
- d. Certain bacteria destroy the activity of penicillin by producing an enzyme penicillinase.
- e. Tuberculosis of cattle, anthrax of sheep is caused by *Mycobacterium tuberculosis* and *Bacillus anthracis* respectively.
- f. Citrus canker by *Xanthomonas citri*, ring rot of potato by *Corynebacterium sepe-donievum* etc.
- g. *Spirochete cytophaga* destroys cotton and article made from it.